

EEE 495 Electrical and Electronics Engineering Design II

Project Specifications



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List of Abbreviations

Abbreviation	Meaning
ADC	Analog-to-Digital Converter
API	Application Programming Interface
CMOS	Complementary Metal–Oxide–Semiconductor
CSV	Comma-Separated Values
DC	Direct Current
DVP	Digital Video Port
EEPROM	Electrically Erasable Programmable Read-Only Memory
ESP32	Espressif Systems 32-bit Wi-Fi/BLE Microcontroller
GB	Gigabyte
GPIO	General Purpose Input/Output
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
I ² C	Inter-Integrated Circuit Bus
IMU	Inertial Measurement Unit
IP	Ingress Protection (environmental sealing rating)
JSON	JavaScript Object Notation
LTE	Long-Term Evolution (4G Cellular Network)
MCU	Microcontroller Unit
mA	Milliampere
MB	Megabyte
MHz	Megahertz
MQTT	Message Queuing Telemetry Transport
NTP	Network Time Protocol
OTA	Over-The-Air Update
PCB	Printed Circuit Board
PIR	Passive Infrared Sensor
PPP	Point-to-Point Protocol
PSU	Power Supply Unit
PWM	Pulse-Width Modulation
RBAC	Role-Based Access Control
RF	Radio Frequency
RSSI	Received Signal Strength Indicator
SVGA	Super Video Graphics Array (800 × 600 px)
TCP/IP	Transmission Control Protocol / Internet Protocol
TLS	Transport Layer Security
UART	Universal Asynchronous Receiver–Transmitter
UI	User Interface
USB	Universal Serial Bus
WAN	Wide Area Network
WAV	Waveform Audio File Format
Wi-Fi	Wireless Fidelity (IEEE 802.11 b/g/n)

1 Introduction

1.1 Brief Description of the Product

FarmGate Sentinel is a vision-first IoT system that detects and deters intrusions near gates, corrals, and fence lines. Edge camera nodes capture baseline images and burst on suspicion; the gateway detects threats and forms local incidents. Then, syncs with the cloud. Users are informed via a mobile app that is connected to the cloud. A simple buzzer turns on to deter threats after the gateway confirms the event. As a secondary objective, the product offers sound-based cognition through a digital microphone; similarly, audio processing occurs on the gateway module. Furthermore, our product offers a monitoring system that detects if the system is broken or misplaced.

1.2 User-Visible Modules (Hardware & Software)

Table 1: System Modules and User Interaction Summary

Module	Type	Purpose	Primary User Interaction
Edge Camera Node	Device	Baseline capture; burst on suspicion; PIR motion; status LED; Reed Switch, IMU system health confirmation.	Install/aim; check LED; On/Off button interaction.
Gateway Unit	Device	Aggregates images/telemetry; local incident logic; burst control; WAN via LTE.	Power/placement; monitor status; minimal handling.
Buzzer	Actuator	Audible on/off tone for deterrence and feedback.	Enable/cancel; subject to duty limits.
Mobile App (UI)	Software	Incident feed, thumbnails, live refresh, acknowledge/comment, buzzer controls; admin policy edit.	View/act on incidents; admins adjust policy.
Gateway	Device	Fast incident proposal, correlation, burst decisions, store&forward during WAN loss.	Indirect; affects responsiveness and bursts.
Cloud Service	Software	Secure messaging/identity, cross-node correlation, object storage, push notifications.	Indirect; delivers alerts, retains media.

1.3 User Characteristics

Table 2: User Classes, Permissions, and Typical Actions in the FarmGate Sentinel System

User Class	Permissions (RBAC)	Typical Actions	Usage
Admin (Farm Owner)	Full: users/roles, policy, actuation, review/audit	Add users; set capture/burst/quiet hours; review incidents/audit logs	Daily
Member (Farm Personnel)	Actuate, acknowledge, view	Open alerts; review thumbnails; trigger/cancel buzzer; add notes	Daily
Installer (Technician)	Provisioning	Perform first-time setup; aim nodes; verify wireless links and connectivity	One-time / rare

1.4 Operational Scenarios / Use Cases

Table 3: System Event Scenarios, Triggers, and User Actions in the FarmGate Sentinel System

Scenario	Trigger	System Response (Key Metrics)	User Action
Predator Alert	Image and sound detection	Gateway proposes incident in ≤ 30 s (p85); node burst ≥ 2 fps for ≥ 10 s; alert delivered in ≤ 10 s (p85).	Operator toggles buzzer, adds note, and acknowledges alert.
Gate Left Open	Sustained state	“open” Informational incident; no default deterrence; UI badge remains visible until gate is closed.	Admin adjusts policy schedule; operator confirms closure.
Gateway Connectivity Outage	Connet- LTE drop at gateway	Capture continues; incidents queued locally; on restoration, backlog drains with $\geq 85\%$ delivery success; notification sent to user.	Operator reviews timeline after recovery and acknowledges.
Edge Connectivity or Sensor Outage	Wi-Fi drop at edge or halted sensor dataflow	Remaining nodes continue capture; affected edge reports outage; notification sent to user.	Operator contacts technician or installer for repair.

1.5 Top Level Architecture

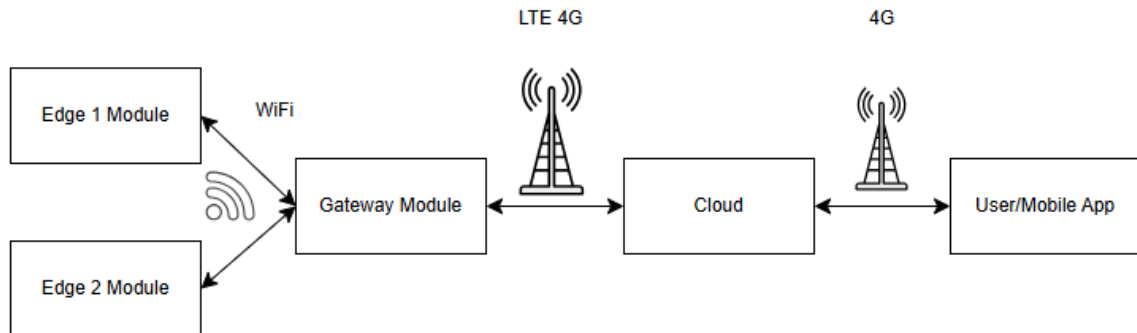


Figure 1: Top Level Architecture

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3 Functional Requirements

In this part, requirements are expressed as measurable, black-box behaviors with concrete verification methods. Each requirement specifies an expected function or performance criterion and includes the method of verification or acceptance testing to ensure compliance during prototype evaluation. Requirements are grouped by subsystem categories: Edge Camera Node, Sensors, Power Unit, Gateway, Communications, Cloud and User Interface to provide clear design objectives and corresponding test procedures. All functions listed herein are considered essential to reliable wildlife detection, timely deterrence activation, and end-to-end data integrity between field nodes, the gateway, and user applications.

Table 4: Functional and Measurable Requirements of the FarmGate Sentinel System

ID	Subsystem / Part	Measurable (Function)	Requirement	Verification / Acceptance Test
2.1 Edge Node				
FR-EDGE-01	Edge Computer	The node shall capture images at a <i>baseline schedule</i> of 10 frames/minute with tolerance ± 1 frame/min, and support a policy-configurable range of 2–30 frames/min without reflash-ing.		Set baseline to 10 fpm; record timestamps for 30 minutes; count frames $\in [290, 310]$. Change policy to 30 fpm; verify new rate takes effect within 60 s and is $30 \pm 5\%$.

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ID	Subsystem / Part	Measurable (Function)	Requirement	Verification / Acceptance Test
FR-EDGE-02	Edge Computer	Default resolution shall be 800×600 (SVGA) with JPEG quality $50\pm5\%$; .High-detail mode will be 1280 ×1024 .In standard mode, 1.44MB will be the average file size		Capture 100 frames per mode; compute mean and 95% file sizes; inspect headers for resolution; verify limits met.
FR-EDGE-03	Sensors	The edge module is able to regulate its integrity. In the event of relocation or replacement, the user will be notified.		We will test this by displacing edge module from a stable edge location. The system will alert if the edge node falls from the determined height.
FR-EDGE-04	Sensors	The system is able to detect objects in front of it in the event of camera obstruction. It will detect objects closer than 1 meter.		We will test this by placing objects at predetermined testing locations, such as from 0 cm to 1m, with incremental step sizes of 10 cm. After 1 meter, we will test in incremental steps of 50 cm.
FR-EDGE-05	Status LED	LED shall indicate states: slow blink (Wi-Fi joining), solid (online), fast blink (burst).		Video analysis of LED over 60 s in each mode; count cycles; verify frequencies within tolerance.
FR-EDGE-06	Actuator	On actuate :buzzer command, an active buzzer shall sound with start latency ≤ 1 s . Default on-time: 20 s ± 1 s .		Command buzzer 10×; measure delay ; verify start latency, and on-time.
FR-EDGE-07	Environmental	Operating temperature -10°C $\rightarrow$ 50°C ; humidity $\leq 95\%$ non-condensing; light rain splash tolerated by enclosure.		Thermal chamber and humidity tests; spray bottle splash; verify continued operation.
2.2 Node Power				
FR-PWR-01	Input range		System operates at 5V DC source. Adapter will be take input 220V DC and convert it to 5V DC for ESP32.	Will be measured with multimeter.
FR-PWR-02	System Fuse		There will be fuses in place to protect the system electronics in the event of an electric spike or overflow, stopping currents larger than 1A.	We will do this by mechanically implementing an overflow using an external power supply.
FR-PWR-03	System Fuse (Reverse / surge)		Node shall survive reverse polarity on input without permanent failure.	Apply reverse polarity for 5 s; verify protection

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ID	Subsystem / Part	Measurable (Function)	Requirement	Verification / Acceptance Test
FR-PWR-04	Average current	At baseline 10 fpm (SVGA), average node current shall be $\leq 360 \text{ mA}$. In idle (no capture, Wi-Fi connected): $\leq 240 \text{ mA}$.		Log current with shunt/DAQ for $\leq 2 \text{ min}$; compute mean; verify thresholds.
2.3 Gateway Node				
FR-GW-01	Detection latency	From image receipt to incident proposal, processing shall be $\leq 2.0 \text{ s}$ (p85) at SVGA.		Feed 25 labeled frames; measure processing time; verify p85 $\leq 2 \text{ s}$.
FR-GW-02	Local buffer	With WAN down, gateway shall buffer $\geq 1000 \text{ images}$ and drain with $\geq 85\%$ success within $\leq 30 \text{ min}$ after link returns.		Disable WAN; collect 1000+ images; restore link; verify drain success and duration.
FR-GW-03	Environmental	Operating temperature $-10^\circ\text{C} \rightarrow 50^\circ\text{C}$; humidity $\leq 95\%$ non-condensing; light rain splash tolerated by enclosure.		Thermal chamber and humidity tests; spray bottle splash; verify continued operation.
2.4 Communications (Node $\leftrightarrow$ Gateway $\leftrightarrow$ Cloud)				
FR-COMM-01	InterSystem Communication	Node-to-gateway communication method is Wi-Fi 2.4GHz, and the gateway communication method is a USB LTE Modem with a SIM card that the user must purchase.		Test will be done by confirming the connection between nodes.
FR-COMM-02	Node Wi-Fi link	Node shall maintain Wi-Fi to the gateway within at least 5m. In the event of a poor connection, the system will switch to a lower-quality image capture.		We will test this by separating the edge and gateway modules from each other with a 50 cm separation per step within the range of [0, 10m].
FR-COMM-03	Node $\rightarrow$ GW latency	Capture-to-gateway receipt latency shall be $\leq 500 \text{ ms}$ (p85) at SVGA; $\leq 1.0 \text{ s}$ (p85) at VGA.		Embed timestamps; compute distributions over 60 frames; verify p85 limits.
FR-COMM-04	Reliability	Successful node $\rightarrow$ gateway delivery shall be $\geq 90\%$ over 5 minutes at 10 fpm.		Count images sent vs. received; verify loss $\leq 1\%$.
2.5 Cloud & Alerting				
FR-CLOUD-01	Correlation	Cloud shall merge detections into incidents when start times are $< 10 \text{ s}$ apart and locations within $\leq 10 \text{ m}$.		Replay two nodes with $\Delta t = 8 \text{ s}$, $\Delta d = 5 \text{ m}$; verify single incident; with $\Delta t = 20 \text{ s}$, verify separate incidents.

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ID	Subsystem / Part	Measurable (Function)	Requirement	Verification / Acceptance Test
FR-CLOUD-02	Storage	For each incident store ≥ 1 thumbnail and ≥ 1 original frame/clip with retention ≥ 7 days .		Create incident; verify objects by key; re-fetch after 7 days in test bucket.
FR-CLOUD-03	Push alerts	On incident confirmation, push notification to registered clients within ≤ 30 s (p85) of gateway detection.		Measure gateway detection time vs. mobile receipt time over 5 trials; verify p85.
FR-CLOUD-04	RBAC	Enforce user roles. Unauthorized actions return HTTP 403 / MQTT auth fail and are audit-logged.		Attempt protected API calls with downgraded tokens; verify 403, MQTT auth failure, and audit log entry.

2.6 User Interface (Web / Mobile)

FR-UI-01	Mobile App Functionalities	UI shall display incident cards with thumbnail , time , node ID , site , controls buzzer, and the mobile app will allow creating, editing, and changing privileges of accounts		Open incident view for test incidents; verify fields and responsive controls. Will test for creating, updating changing user roles
FR-UI-02	Live view	On request, UI shall fetch latest available image per node within ≤ 10 s (p95).		Click “refresh” 5 times; measure server time to image render; verify p95 ≤ 10 s.
FR-UI-03	Policy edit	Admin can update per-node schedule (fpm), burst thresholds, and actuator durations; changes take effect within ≤ 60 s.		Change policy; observe behavior shift at node/gateway; verify ≤ 60 s propagation.

4 Non-functional Requirements / Specifications

Temperature ambient determined as $20 \pm 5^\circ\text{C}$; supply at nominal; values are for a two-edge-node deployment. Items are categorized as follows: Verifiable by Test (T), Demonstration (D), Analysis (A), or Inspection (I).

4.1 Size and Weight of Physical Sub-Components

The physical characteristics of each main hardware component have been considered to ensure portability, ease of installation, and mechanical safety under field conditions. The approximate dimensions and weights are summarized in Table 5.

Table 5: Approximate Size and Weight of Physical Sub-Components

Component	Approx. Size (mm)	Approx. Weight (g)	Remarks
Gateway Computer	85 × 56 × 18	46	Mounted inside the gateway enclosure; compact and lightweight for pole or wall installation.
LTE Modem (USB Stick)	90 × 27 × 12	35	Adds minimal mass; connects via USB-A port.
microSD Card (32 GB)	15 × 11 × 1	2	Negligible effect on overall size and weight.
Edge Module	100 × 80 × 50	500	Weather-resistant PLA/ABS; protects internal components.
Gateway Computer Adapter	70 × 30 × 90	130	Primary power source for Raspberry Pi.
Gateway Enclosure	150 × 110 × 70	100	Mounted near fence or shed; total portable weight.

- Each **edge node** weighs approximately **500–550 g** enclosure.
- The **gateway unit** weighs around **200-250 g**, suitable for mounting on a fixed positions.
- The total deployed system (gateway + two edge nodes) remains under **1.5 kg**, ensuring portability and low-impact mounting in rural environments.

4.2 Power Supply Specifications

Each subsystem of the FarmGate Sentinel has distinct power sources and tolerances. The gateway operates continuously from a regulated adapter, whereas the edge nodes are battery-powered for autonomous field operation. Table 6 summarizes the voltage, current, and supply characteristics for all key components.

Table 6: Power Supply Specifications of Each Sub-Component

Component	Power Source	Operating Voltage (V)	Current (mA)	Remarks
Gateway Computer	5 V DC Adapter (USB-C)	5.0 ± 0.25	3000	Requires official 5 V 3 A PSU for stable operation under CPU and USB load.
LTE Modem (USB Stick)	USB port of Pi (plugged)	5.0	500–700	Powered directly through the Pi; may cause reboot if $PSU < 3$ A.
microSD Card (32 GB)	Internal Pi slot	3.3	<100	Draw supplied by Pi's internal regulator.
Edge Computer	Edge Computer PSU 3.3V DC Adaptor	3.3	160 avg / 400 peak	MP1584 converter provides regulated 3.3 V from ~6 V input.
Edge Sensors	Sensors connected to each edge	3.3	20–120	Negligible draw compared to ESP32 main load.
Gateway PSU (5 V 3 A)	AC mains adapter (220 V $\rightarrow$ 5 V)	5.1 regulated	regu- 3000 max	Official PSU ensures stable 5 V rail for gateway and connected USB devices.
Edge PSU (3.3 V 500 mA)	AC mains adapter (220 V $\rightarrow$ 3.3 V)	3.3 regulated	regu- 500 max	Handmade PSU ensures stable 3.3 V rail for edge and connected USB devices.

- The **gateway** is continuously powered from an official 5 V 3 A USB-C adapter to ensure stable operation under high load.
- Voltage tolerances are maintained within ± 5 %, protecting Edge Computer and sensor circuitry.
- Average edge-node power consumption is below 0.7 W, allowing extended duty cycles with sleep modes.

4.3 Electrical and Electronics Hardware Interface Specifications

The FarmGate Sentinel system integrates multiple wired and wireless interfaces between its gateway and edge-node subsystems to enable seamless data transfer, sensing, and control. The following specifications summarize the hardware interfaces available to both the user and the system components in table Table 7.

Table 7: Electrical and Electronic Interface Specifications

Interface Type	Component(s)	Electrical Level / Standard	Direction	Function / Description
USB 2.0 Port	Gateway Computer 5 ↔ LTE Modem	5 V / USB 2.0	Bi direction	Provides internet backhaul via LTE modem; data and power shared over a single interface.
USB 3.0 Port	Gateway Computer (peripheral expansion)	5 V / USB 3.0 (Blue)	Bi direction	Allows connection of storage or diagnostic peripherals during maintenance.
microSD Card Interface	Gateway Computer / Edge Computer	3.3 V logic (SPI)	Bi direction	Local storage of firmware, images, and configuration data.
Wi-Fi (IEEE 802.11 b/g/n)	Edge Computer ↔ Gateway Computer	2.4 GHz band	Bi direction	Primary wireless communication channel for audio, video, and sensor data transmission.
GPIO Pins	Edge Node	3.3 V TTL	I/O	Controls sensors (PIR, reed) and actuators (buzzer, LED); protected by series resistors.
I ² C Bus	Edge Computer ↔ IMU Sensor	3.3 V logic (SCL, SDA)	Bi direction	Transfers motion and orientation data from IMU module to MCU.
Analog Microphone Input	Digital Microphone Module → Edge Computer	3.3 V bias / AC-coupled	Input	Captures environmental audio for sound-based detection.
Camera Interface (OV2640)	Edge Computer	3.3 V parallel DVP	Input	Provides still and streaming image data for wildlife recognition.
Alarm Output (Buzzer + LED)	Edge GPIO → Actuators	3.3 V logic / ≤10 mA	Output	Triggers local audible and visual deterrent signals.
Power Input Port	Gateway Computer (USB-C) / Edge Node (2-pin terminal)	5 V DC / 3.3 V DC	Input	External power from PSU (gateway) or buck converter (edge nodes).

4.4 Software Interface Specifications

The FarmGate Sentinel system employs multiple software-level communication interfaces between the gateway, edge nodes, and cloud server. These interfaces ensure reliable data transfer, device coordination, and system interoperability. Table 8 summarizes the key software and communication protocols utilized in the system.

Table 8: Software Interface and Communication Specifications

Interface / Protocol	Used Between	Type	Description / Functionality
Wi-Fi Communication (IEEE 802.11 b/g/n)	Edge Module ↔ Gateway Module	Wireless (TCP/IP)	Primary communication link for transmitting images, audio data, and sensor readings. Each node connects to a secure TCP socket or HTTP REST endpoint hosted on the gateway.
HTTP / REST API	Gateway Server ↔ Cloud	Application Layer (over LTE)	Used for event uploads and notifications. Encoded data (JPEG, WAV, JSON) is sent to cloud endpoints for storage and retrieval by the mobile app.
MQTT	Gateway ↔ Cloud	Publish–Subscribe	Supports lightweight telemetry, topic-based detection updates, and real-time alerts between gateway and cloud.
USB Serial Interface	Gateway ↔ Computer / LTE Modem	Wired (USB CDC / PPP)	Provides LTE network connectivity using the Point-to-Point Protocol (PPP) over USB.
UART (Serial Communication)	Inside Edge Module	Wired (3.3 V TTL)	Used for debugging, configuration, or firmware updates via serial console (115200 baud, 8N1).
I ² C Data Bus Protocol	Inside Edge Module	Embedded Bus	Transfers motion and orientation data using a standard 100 kHz clock and 7-bit addressing.
Data Formats	All Subsystems	JSON / CSV / Binary	Images stored as JPEG, audio as WAV, and sensor data as JSON or CSV for logging and cloud upload.
Wi-Fi Security	Edge to Gateway ↔ Gateway	WPA2-PSK	Provides encrypted transmission of multimedia and sensor data between nodes and gateway.
LTE Network Protocols	Gateway ↔ Cloud	IPv4 / TCP	Cloud data is transmitted using standard IP-based protocols via mobile broadband.

4.5 User Interface in Engineering Terms

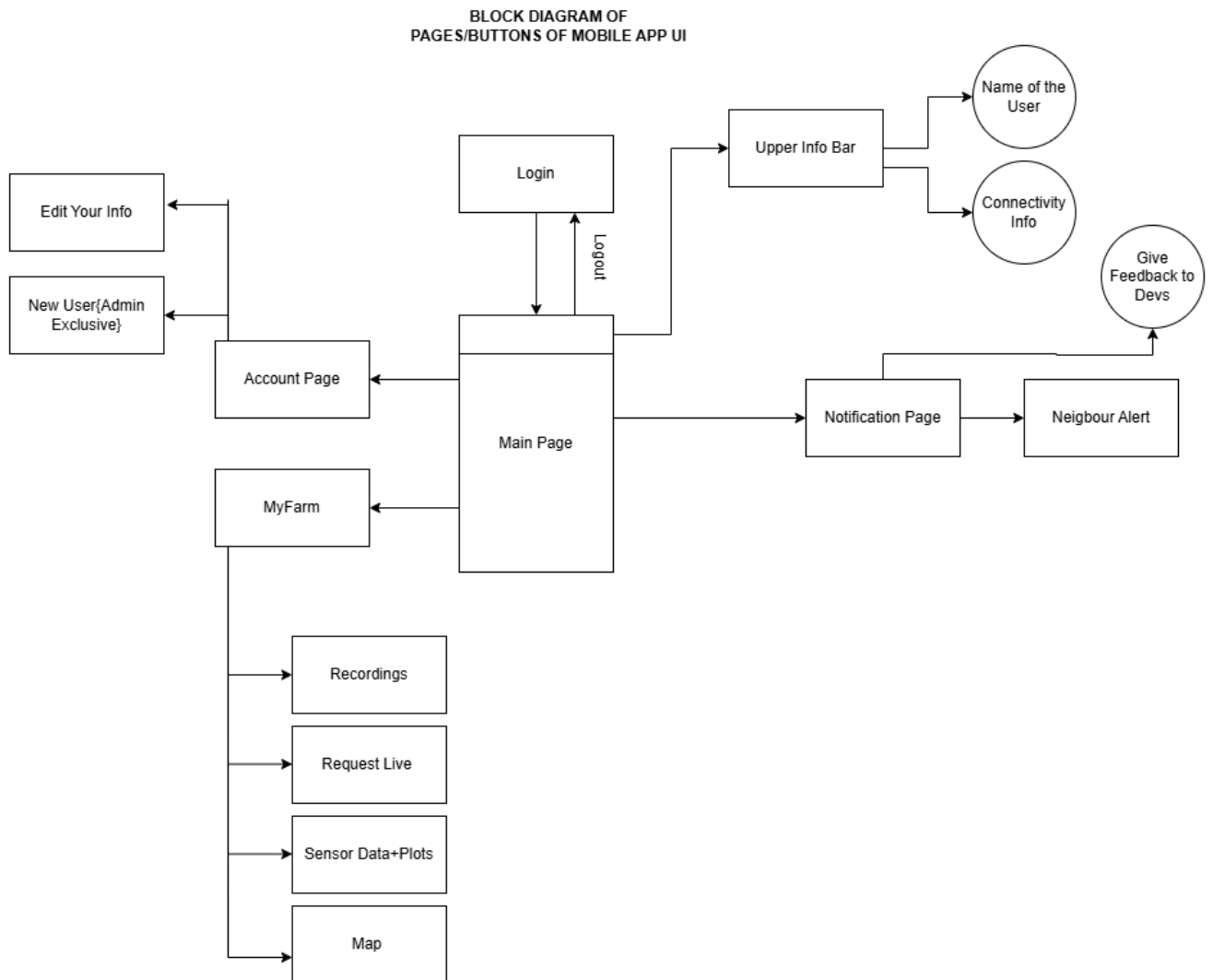


Figure 2: Block Diagram of of User Interface

The FarmGate Sentinel mobile application provides the primary user interface for farmers and cooperative users to monitor wildlife activity, receive alerts, and view captured evidence. The interface design emphasizes functionality, reliability, and clarity, enabling users to interact with the cloud system and gateway in real time. Table 9 summarizes the user types, screen layout, and main functions in engineering terms.

Table 9: User Interface Specification for the Mobile Application (Revised)

Screen / Module	User Type	Access Method	Main Functionalities and Controls
Login Screen	All Users	App Launch	• Secure login using ID and password. • Redirects to Main Page upon successful authentication. • Includes password recovery and “New Member” registration option.
Main Page	All Users	After Login	• Central hub linking all modules. • Access to all other pages.
My Farm Page	All Users	From Main Page	• Displays real-time data and farm-related modules. • Provides access to recordings, and Request Live.
Map Page	All Users	From My Farm	• Displays location of sensors and cameras. • Allows zoom and node-specific data view.
Recordings Page	All Users	From My Farm	• Shows stored videos and detections. • Allows playback, delete, or download from cloud.
Sensor Data + Plots Page	All Users	From My Farm	• Plots temperature, motion, and sound data. • Enables historical data visualization.
Request Live Page	All Users	From My Farm	• Initiates live stream from cameras. • Shows real-time video feed with audio.

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Screen / Module	User Type	Access Method	Main Functionalities and Controls
Notification Page	All Users	From Main Page	• Lists all recent alerts and notifications. • Links to Neighbor Alert page. • Allows filtering by date or event type.
Neighbor Alert Page	Neighboring Farm Users	From Notification Page	• Displays alerts shared by nearby farms. • Provides feedback and acknowledgment options.
Feedback Module	All Users	From Notifications / Alerts	• Allows users to send feedback or report issues. • Supports quick responses to alerts.
Account Page	All Users	From Main Page	• Displays user profile and device details. • Provides access to Edit Your Info and New Member setup. • Includes logout option.
Edit Your Info Page	All Users	From Account Page	• Modify account information such as name, password, and email.
New Member Registration Page	Farm Owner	From Account Page / Login	• Register a new user or farm. • Stores data securely to cloud database.
Top Info Bar	All Users	Appears Globally	• Displays app name, network status, and notification icon. • Allows quick navigation to Notification Page.
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Screen / Module	User Type	Access Method	Main Functionalities and Controls
Logout Function	All Users	From Account or Main Page	• Safely terminates session. • Redirects to Login Screen.

4.6 Environmental Operating Conditions

The FarmGate Sentinel system is designed for outdoor agricultural environments where temperature, humidity, and exposure to dust or rain may vary significantly. Both the gateway and edge nodes are enclosed in weather-protected casings to ensure reliable operation under field conditions. The environmental specifications are summarized in Table 10.

Table 10: Environmental Operating Conditions of System Components

Parameter	Gateway	Edge Node	Remarks
Operating Temperature Range	0–50 °C	–10–60 °C	Edge node enclosure includes passive ventilation; system components selected for wide temperature tolerance.
Storage Temperature Range	–20–70 °C	–20–70 °C	All components can be safely stored in unpowered state within this range.
Ingress Protection (en-closure rating)	Indoor	Outdoor (3D-printed ABS housing)	Protects against dust and low-pressure water jets; ensures rain and debris resistance.
Vibration Tolerance	Moderate (fixed installation)	Moderate (fence or pole mounting)	Mechanical fastening minimizes vibration effects from wind or nearby machinery.
Operational Conditions	Weather Protected (gateway box)	Outdoor (fence, post, or shelter)	System designed to function in rain, fog, and moderate wind conditions.